

Procedure to assemble the OptoGrid device

Ver.	Author	Date	Note
1.0	Fukui (Daniel) Yang, Sainsbury Wellcome Centre, UCL	14/01/2026	Initial Commit
1.1	Fukui (Daniel) Yang	23/01/2026	Changed 3D model for head-fixed base plate; magnets from N42 to N52

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Module 1: Printed Circuit Board and Assembly

Step 1: Order Main PCBA, Battery PCBA, and uLED matrix PCBA

1. There are three PCBA components to the OptoGrid device, they can be ordered from commercial PCBA manufacturer with the production files and specification below:

	PCBA File Location	Recommended Manufacturer	PCB Production Specification
1	PCB/OptoGrid_V0.05/OptoGrid_Battery_V0.05	PCBWay	PCB Type: Standard PCB PCB Thickness: 0.6mm Layer: 2 Layers Material: FR-4 Surface finish: HASL lead free Silkscreen: White Min hole size: 0.3mm diameter Min track/spacing: 4/4mil Via process: Tenting vias Finished copper: 1 oz Cu Solder mask: Green
2	PCB/OptoGrid_V0.05/OptoGrid_Main_V0.05		
3	PCB/OptoGrid_V0.05/OptoGrid_uLEDArray_V0.05		PCB Type: Flexible PCB FPC Thickness: 0.1mm Layer: 2 Layers Material: Polyimide Flex Surface finish: Immersion gold (ENIG), 1U Silkscreen: None Min hole size: 0.15/0.35mm diameter Min track/spacing: ≥0.06mm Via process: N/A Finished copper: 1 oz Cu Solder mask: Yellow Coverlay

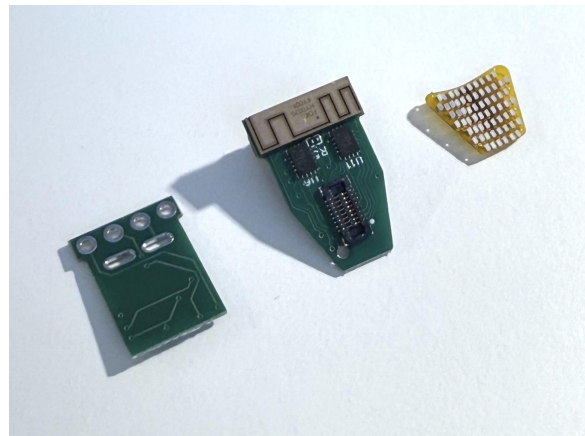
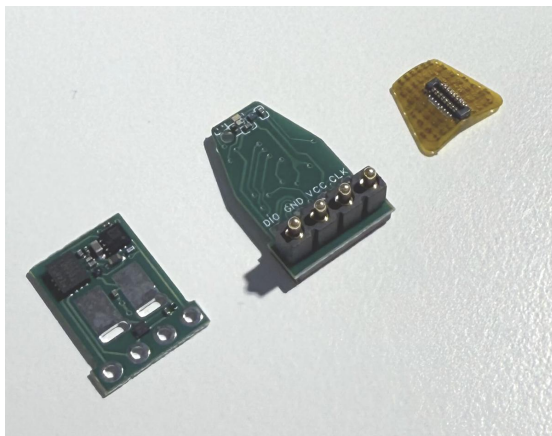
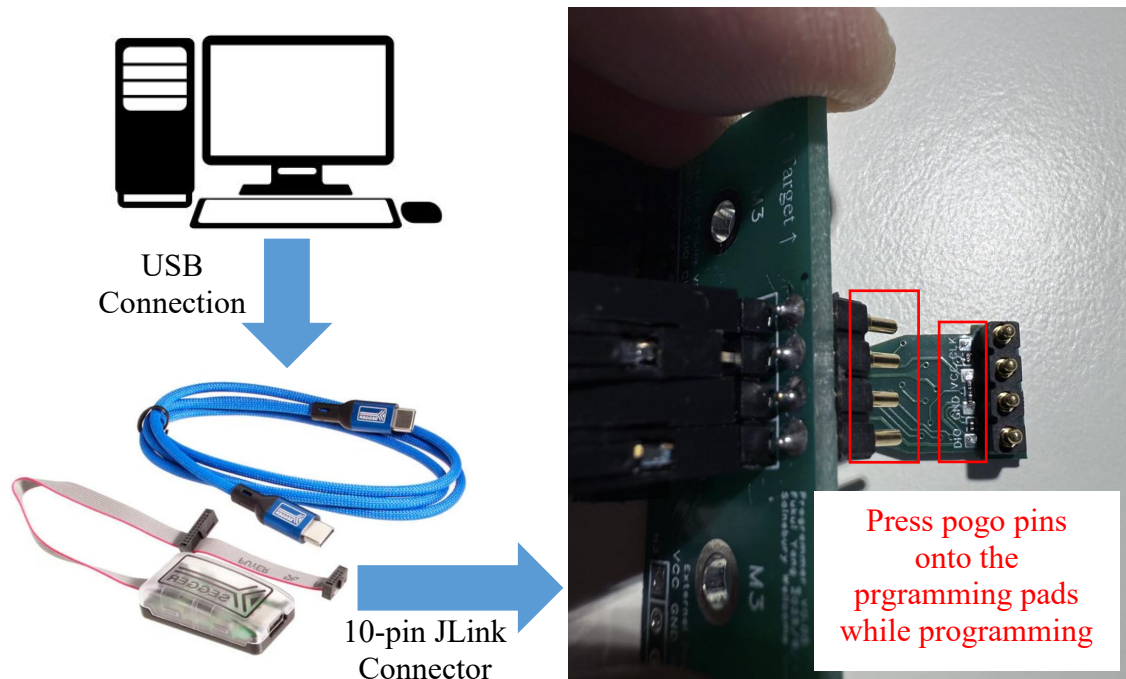


Figure 1: Photos of the three PCBA components that arrives from the manufacturer

Step 2: Flash firmware onto the Main PCBA

1. Use [Programmer_V0.05 PCBA] and a JLink (e.g. JLink mini edu) to flash the firmware.



Note: To match 10-pin JLink connector orientation, use pin 1 marked by two PCB's silkscreen

2. Use the [Programmer] functionality from [NRF Connect Software](#) to perform the flashing.

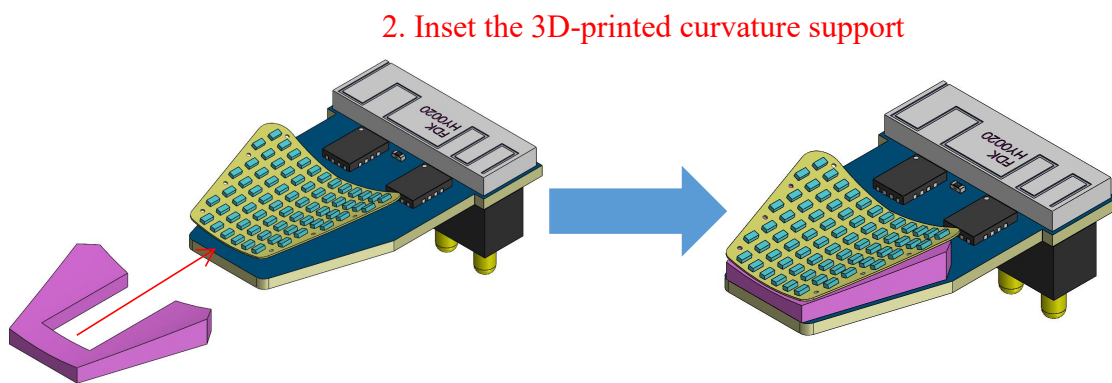
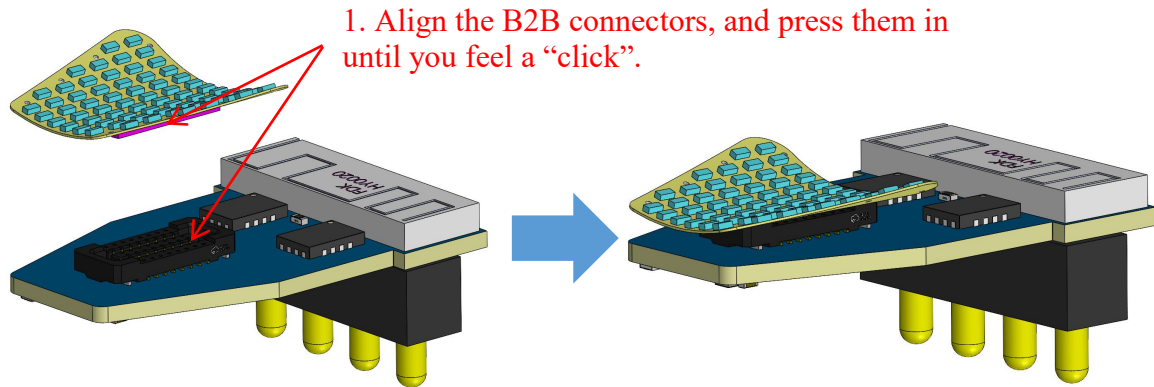
2. Select the JLink device, while it is pressed onto the Main PCBA

1. Erase & Write

3. Add firmware's 3 hex files:

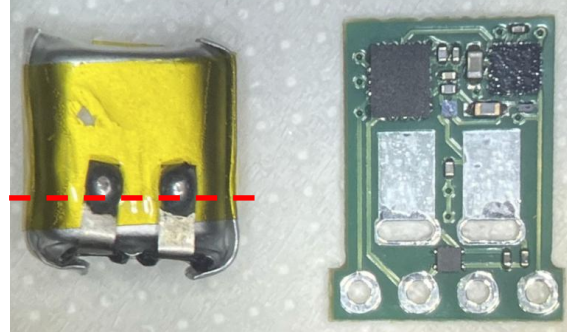
The screenshot shows the Programmer v4.6.0 software interface. The left sidebar contains a 'FILE' section with 'Add file', 'Reload files', and 'Clear files' buttons. Below this is a 'DEVICE' section with 'Erase all', 'Erase & write', 'Save as file', 'Reset', 'Write', and 'Read' buttons. The 'J-LINK SETTINGS' section has 'Auto read memory' and 'Auto reset' checkboxes. The 'MCUBOOT SETTINGS' section has a 'SHOW SIDE PANEL' checkbox. The main area shows a 'File memory layout' and a 'Device memory layout'. A red arrow points from the 'Add file' button to a list of three hex files: 'E:\tech-dev\OptoGrid\OptoGrid_Firmware_Signed\softdevice_s132.hex', 'E:\tech-dev\OptoGrid\OptoGrid_Firmware_Signed\secure_bootloader.hex', and 'E:\tech-dev\OptoGrid\OptoGrid_Firmware_Signed\nrf52832_xxaa.hex'. A red arrow points from the 'Erase & write' button to the 'Erase & write' button in the 'DEVICE' section. A red arrow points from the 'Add file' button to the 'Add file' button in the 'FILE' section.

Step 3: Assemble the uLED matrix to the Main PCBA



Step 4: Solder battery onto the battery PCBA

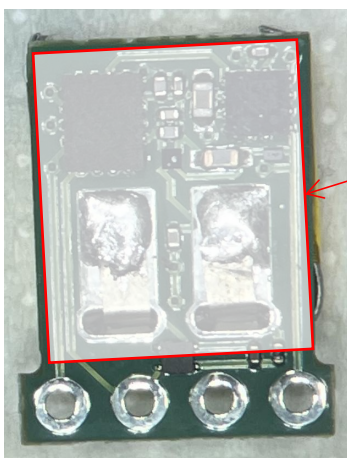
1. Flip the leads up to be perpendicular to the battery's surface, and cut away excess leads (part of the lead above dashed line) from the battery.



2. Insert the two leads through the holes on the PCBA, and flip the leads onto the solder pads, then solder the leads to the pads (solder tip temperature: 320 degree C)



3. Apply and cure UV glue (e.g. RelyX Unicem 2 Automix, 3M) to cover the solder points and electronics



Region to cover with glue to offer insulation and protection



Module 2: Housing Unit Assembly

Step 1: Order 3D printing of housing units

1. 3D print the 5 following parts per OptoGrid device.

- a) Recommended printer: [Form 4B \(FormLabs\)](#)
- b) Recommended resin: [Labeled below using resin marketed by FormLabs](#)

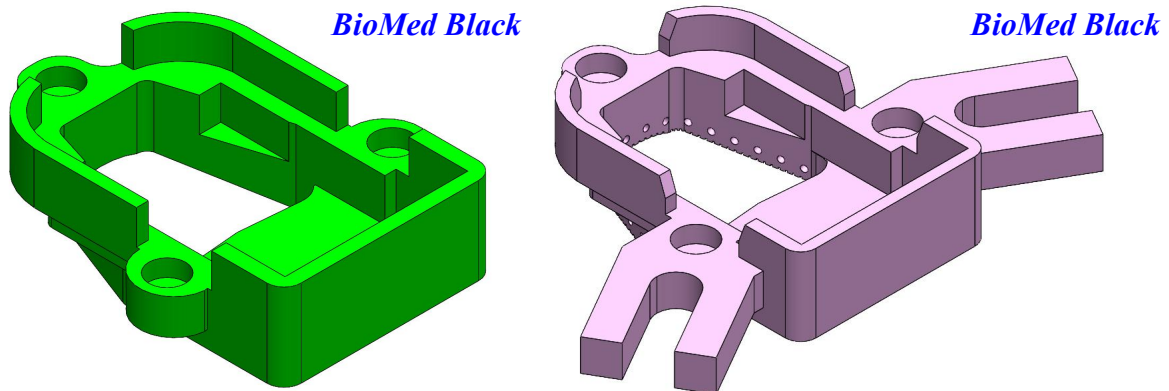


Figure 2: Lower housing units (Left: For non-headfixed, Right: for headfixed)

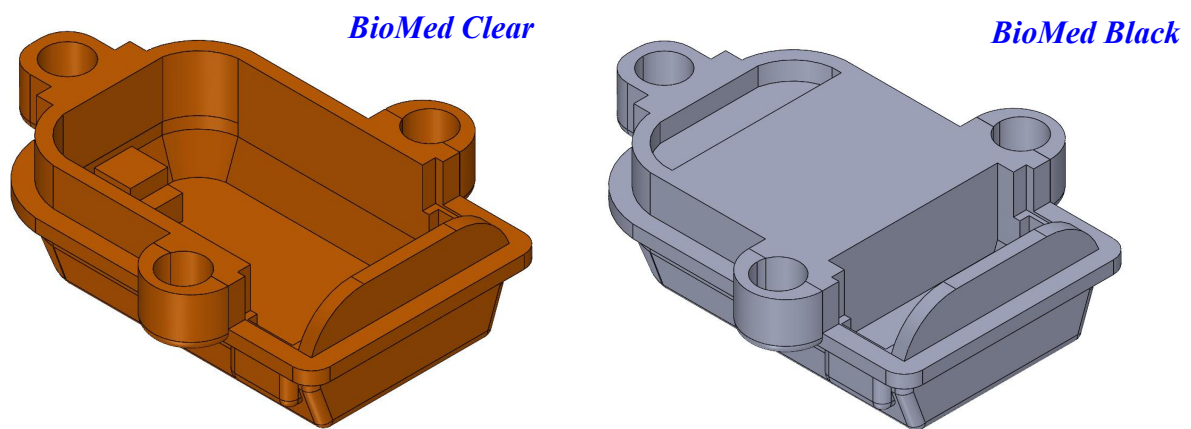


Figure 3: Upper housing units (Left: Battery Cap, Right: Sham Cap)

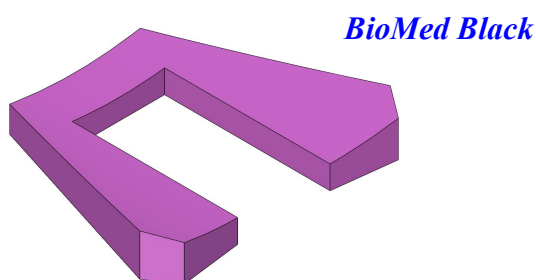
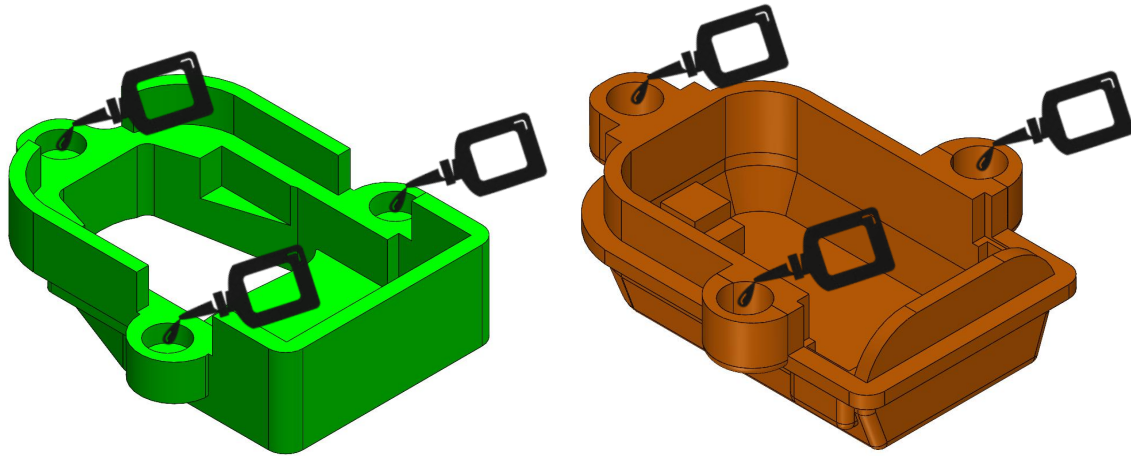


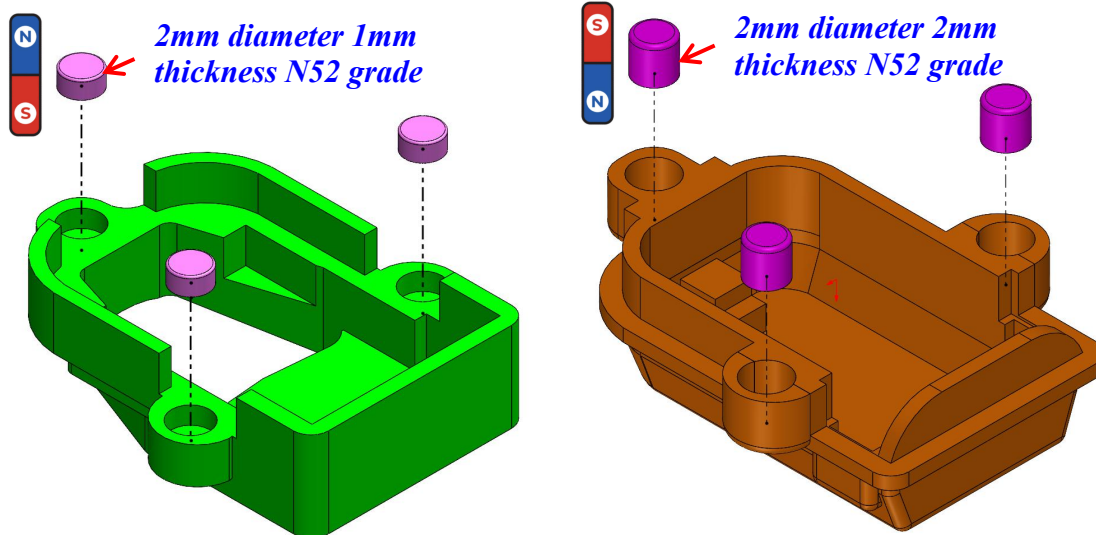
Figure 4: uLED matrix curvature support

Step 2: Glue magnets to the housing units

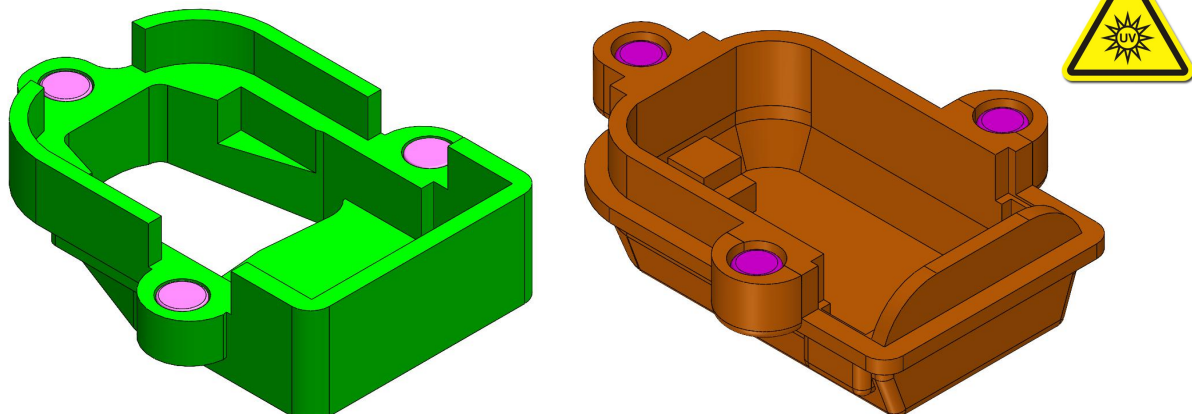
1. Apply UV-curable glue into the holes (e.g. RelyX Unicem 2 Automix, 3M)



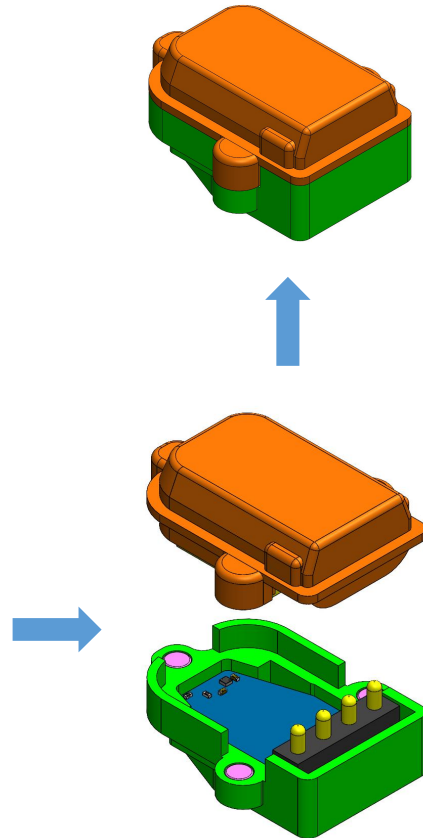
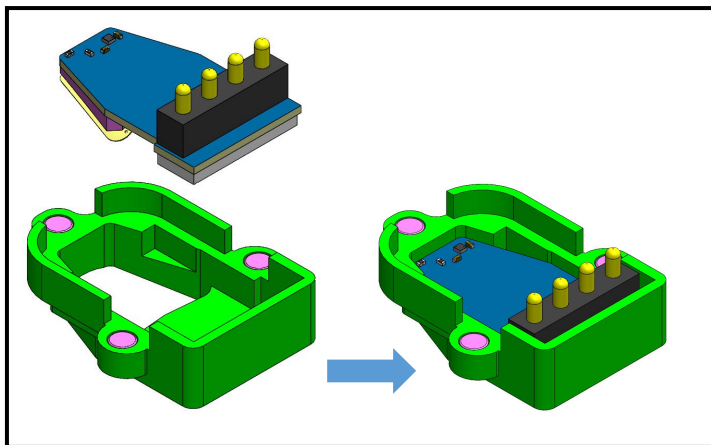
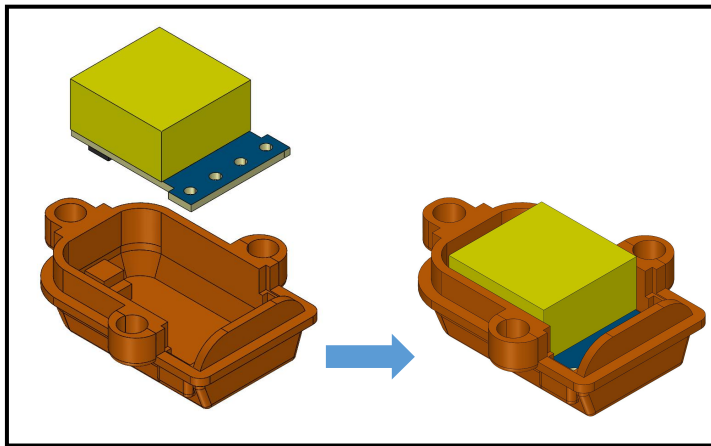
2. Insert the magnet into the holes



3. UV curate the glue (20 second minimum)



Step 3: Test assembly of the PCBA with the housing units



Module 3: Software Deployment

1. Please refer to this GitHub repository for deployment of the PC software that interfaces with the OptoGrid device. <https://github.com/danielfrankch/optogrid-pc-app>

2. Quick guide on how to use the web-based GUI to interact with the OptoGrid device.

The screenshot shows the 'OptoGrid BLE Browser' web application running in a browser at localhost:3000. The interface includes a top navigation bar, a central control area with buttons for 'Scan', 'Select device...', 'Connect', and 'Debug Mode', a log window showing device status, a 'Trigger' button, and a 'Parameter Table' for programming the device. On the right, there's a 'Toggle the LEDs to select' brainmap, a 'SHAM'/'IMU'/'STATUS' section with checkboxes for 'uLED Scan', 'Last Stim', and 'Read Battery', and a 'IMU Data (last 200 samples)' section with 'Accelerometer (g)' and 'Gyroscope (°/s)' plots.

1) Scan → Click the 'Scan' button.

2) Select your OptoGrid device → Click the 'Select device...' dropdown menu.

3) Connect → Click the 'Connect' button.

4) These toggles the on/off of:
 - Sham LED (Blue)
 - Status LED (Red)

5) Click to trigger the current programmed stimulation → Click the 'Trigger' button.

6) Toggle the LEDs here in this brainmap to intuitively select your LEDs → Click on the LED map in the brainmap.

6a) Input your desired parameter value in this column → Input the value in the 'Write Value' column of the Parameter Table.

7) Click to program the current Parameter Table to the device → Click the 'Write' button.

8) Try the debug functions:
 - uLED Scan: Scans all 64 LEDs, and the broken LEDs will be marked with a red X in the LED map above
 - Last Stim: Prints the last triggered stimulation's timestamp
 - Read Battery: Updates to the latest battery voltage value

9) Toggle IMU recording → Click the 'IMU' checkbox. Recorded data can be found in the folder named "data"

Parameter	Value	Write Value	Unit
Sequence Length	1	1	count
LED Selection	18446744073709551615	18446744073709551615	bitmap
Duration	550	550	ms
Period	100	100	ms
Pulse Width	1	1	ms
Amplitude	100	100	%
PWM Frequency	50000	50000	Hz
Ramp Up Time	0	0	ms
Ramp Down Time	0	0	ms